

Lab Assignment 21.3 – AI Assisted Coding

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Lab Assignment 21.3 – Collaborative AI Coding

Task 1: AI-Assisted Pair Programming (Expense Tracker)

Question

Develop an Expense Tracker using AI and improve it with additional features.

Prompt

"Generate a Python expense tracker program and enhance it with input validation, monthly summary, and modular functions."

Code

Initial Version

```
expenses = []

while True:

    name = input("Enter expense name: ")
    amount = int(input("Enter amount: "))
    expenses.append((name, amount))

    choice = input("Add more? (y/n): ")
    if choice == 'n':
        break

print(expenses)
```

Improved Version

```
def add_expense(expenses):
    name = input("Enter expense name: ")
```

```
try:
    amount = float(input("Enter amount: "))
except:
    print("Invalid input")
    return
expenses.append((name, amount))

def show_summary(expenses):
    total = sum(amount for _, amount in expenses)
    print("Total Expense:", total)

expenses = []

while True:
    add_expense(expenses)
    choice = input("Add more? (y/n): ")
    if choice == 'n':
        break

show_summary(expenses)
```

Output

```
Enter expense name: Food
Enter amount: 200
Add more? n
Total Expense: 200.0
```

Code Explanation

- Initial version stores expenses simply
- Improved version uses functions
- Adds input validation

- Calculates total expense (monthly summary)
-

Task 2: Version Control Integration (Student Grade Calculator)

Question

Implement Git workflow for a student grade calculator project.

Prompt

"Create a Python program for grade calculation and describe Git branching, commits, and merging."

Code

```
marks = [80, 75, 90]
```

```
avg = sum(marks)/len(marks)
```

```
if avg >= 90:
```

```
    grade = 'A'
```

```
elif avg >= 75:
```

```
    grade = 'B'
```

```
else:
```

```
    grade = 'C'
```

```
print("Average:", avg)
```

```
print("Grade:", grade)
```

Output

Average: 81.66

Grade: B

Code Explanation

- Calculates average marks

- Assigns grade based on condition
- Demonstrates feature added in branch

Git Steps

git init

git add .

git commit -m "Initial commit: basic grade calculator"

git branch feature-grade

git checkout feature-grade

add grade logic

git commit -m "Added grade classification feature"

git checkout main

git merge feature-grade

Task 3: AI for Commit Message Generation

Question

Generate commit messages for password validation improvements.

Prompt

"Generate meaningful Git commit messages for adding password validation rules."

Code

```
import re
```

```
def validate(password):
```

```
    return len(password) >= 8 and re.search(r"[!@#$]", password)
```

```
print(validate("Test@123"))
```

Output

True

Code Explanation

- Checks minimum length
 - Ensures special character
 - Improves password security
-

AI Generated Commit Messages

"Added password length validation"

"Implemented special character check for stronger passwords"

"Improved security in password validation module"

Task 4: Pair Programming Simulation (Quiz System)

Question

Develop a quiz system collaboratively with added features.

Prompt

"Create a Python quiz program and enhance it with score calculation and validation."

Code

Initial Version (Student A)

```
questions = {  
    "2+2": "4",  
    "3+3": "6"  
}
```

for q in questions:

```
ans = input(q + ": ")
if ans == questions[q]:
    print("Correct")
else:
    print("Wrong")
```

Improved Version (Student B + AI)

```
questions = {
    "2+2": "4",
    "3+3": "6"
}

score = 0

for q in questions:
    ans = input(q + ": ")
    if ans.strip() == questions[q]:
        print("Correct")
        score += 1
    else:
        print("Wrong")

print("Final Score:", score)
```

Output

```
2+2: 4
Correct
3+3: 6
Correct
Final Score: 2
```

Code Explanation

- Initial version checks answers
 - Improved version adds score tracking
 - Input validation using `strip()`
 - Demonstrates collaboration
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